

Linux terminal

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Comandos

Basic Linux Commands

1. **pwd** — Prints the current working directory you're in.
2. **ls** — Lists the files and directories in the current folder. .
3. **cd** — Changes the directory you're working in.
4. **clear** — Clears the terminal screen.
5. **echo** — Displays a line of text or variable value.
6. **exit** — Closes the terminal session.
7. **history** — Displays a list of previously used commands. .
8. **man** — Opens the manual page for a command.
9. **whoami** — Prints the current logged-in username.
10. **hostname** — Shows the system's hostname.

File and Directory Operations 11.touch – Creates a new empty file.

12. **mkdir** — Creates a new directory.
13. **rmdir** — Removes empty directories.
14. **rm** — Removes files or directories.
15. **cp** — Copies files or directories.
16. **mv** — Moves or renames files.
17. **stat** — Displays detailed file information.
18. **file** — Identifies the type of file (text, binary, image, etc.).
19. **basename** — Returns the file name from a full path.
20. **dirname** — Returns the directory path from a full file path.

File Permissions & Ownership

21. **chmod** — Changes file or directory permissions.
22. **chown** — Changes file ownership (user and/or group).
23. **chgrp** — Changes the group ownership of a file.
24. **umask** — Sets default permissions for newly created files.
25. **lsattr** — Lists file attributes on a Linux file system.
26. **chattr** — Changes file attributes on a Linux file system.
27. **getfacl** — Displays Access Control List (ACL) of a file.
28. **setfacl** — Sets ACL permissions on files or directories.

User & Group Management

29. **adduser** — Adds a new user to the system.
30. **useradd** — Low-level utility to add a user.
31. **passwd** — Changes user password.
32. **usermod** — Modifies a user account.
33. **userdel** — Deletes a user account.
34. **groupadd** — Creates a new group.
35. **groupdel** — Deletes a group.
36. **groupmod** — Modifies an existing group.
37. **id** — Displays user ID and group ID.
38. **who** — Shows who is logged in.
39. **w** — Shows logged-in users and what they are doing.

- 40. **groups** – Displays groups a user belongs to.
- 41. **su** – Switches to another user account.
- 42. **sudo** – Executes a command with elevated privileges.

Process Management

- 43. **ps** – Shows running processes.
- 44. **top** – Displays real-time system processes and usage.
- 45. **htop** – Interactive process viewer (enhanced top).
- 46. **kill** – Sends signals to processes (usually to terminate).
- 47. **killall** – Sends signals to all processes by name.
- 48. **nice** – Starts a process with a given priority.
- 49. **renice** – Changes priority of a running process.
- 50. **bg** – Resumes a job in the background.
- 51. **fg** – Brings a background job to the foreground.
- 52. **jobs** – Lists active jobs in the shell.
- 53. **pidof** – Finds the process ID of a running program.
- 54. **watch** – Repeats a command periodically and shows output.

Disk & Filesystem Commands

- 55. **df** – Reports file system disk space usage.
- 56. **du** – Estimates file or directory space usage.
- 57. **mount** – Mounts a file system.
- 58. **umount** – Unmounts a file system.
- 59. **fsck** – Checks and repairs file systems.
- 60. **blkid** – Displays block device information.
- 61. **lsblk** – Lists block devices in a tree-like structure.
- 62. **parted** – Manages disk partitions interactively.
- 63. **fdisk** – Partition table manipulator for Linux.
- 64. **mkfs** – Creates a new file system.
- 65. **tune2fs** – Tunes file system parameters.
- 66. **e2label** – Changes the label of an ext2/ext3/ext4 filesystem.

Archiving & Compression

- 67. **tar** – Archives files into **.tar** format.
- 68. **gzip** – Compresses files using **.gz** format.
- 69. **gunzip** – Decompresses **.gz** files.
- 70. **bzip2** – Compresses files using **.bz2** format.
- 71. **bunzip2** – Decompresses **.bz2** files.
- 72. **xz** – Compresses files using **.xz** format.
- 73. **unxz** – Decompresses **.xz** files.
- 74. **zip** – Compresses files into **.zip** archive.
 - 1. **zip arquivo.zip arquivo.md**
- 75. **unzip** – Extracts files from **.zip** archives.
 - 1. **unzip -l arquivo.zip**: lista os arquivos dentro do **.zip**.
 - 1. **-l**: list
 - 2. **-r**: recursive
- 76. **7z** – High-compression archiver for **.7z** files.

77. **zcat** – Views contents of a compressed file.

Networking Commands

- 78. **ip** – Configures and displays IP networking.
- 79. **ifconfig** – Displays or configures network interfaces (deprecated but still used).
- 80. **ip a** – Displays all network addresses (modern alternative to ifconfig).
- 81. **ping** – Checks connectivity to another host.
- 82. **traceroute** – Traces the route packets take to a host.
- 83. **netstat** – Displays network connections, routing tables, and stats (older tool).
- 84. **ss** – Displays detailed socket statistics (modern replacement for netstat).
- 85. **dig** – Queries DNS name servers.
- 86. **nslookup** – Performs DNS lookups (older tool).
- 87. **host** – Simple DNS query tool.
- 88. **curl** – Transfers data from or to a server using supported protocols.
- 89. **wget** – Non-interactive network downloader.
- 90. **telnet** – Connects to remote machines using Telnet protocol.
- 91. **ssh** – Connects to remote machines securely.
- 92. **scp** – Securely copies files between systems.
- 93. **rsync** – Efficiently syncs files and directories between systems.
- 94. **ftp** – Transfers files over FTP (less secure, older protocol).
- 95. **nmcli** – Command-line tool for controlling NetworkManager.
- 96. **nmap** – Network scanner for hosts and open ports.
- 97. **tcpdump** – Captures and analyzes network packets.
- 98. **iptables** – Manages firewall rules.

Package Management (Debian/Ubuntu)

- 99. **apt** – Modern package management tool for Debian-based systems.
- 100. **apt-get** – Legacy tool for package operations.
- 101. **apt-cache** – Queries package information.
- 102. **dpkg** – Low-level Debian package management tool.
- 103. **snap** – Manages snap packages (universal packages).
- 104. **update-alternatives** – Manages default system applications.

Package Management (RHEL/CentOS/Fedora)

- 105. **yum** – Package manager for RPM-based systems (older systems).
- 106. **dnf** – Modern replacement for yum.
- 107. **rpm** – Low-level RPM package management.
- 108. **repoquery** – Queries repository information.
- 109. **dnf info** – Retrieves package info from DNF repos.
- 110. **dnf clean** – Clears metadata cache.

System Information

- 111. **uname** – Displays system information like kernel version.
- 112. **hostnamectl** – Controls system hostname and related settings.
- 113. **uptime** – Shows how long the system has been running.
- 114. **whoami** – Displays the current username.
- 115. **id** – Shows user ID and group ID.

- 116. **top** – Displays dynamic real-time view of running processes.
- 117. **vmstat** – Reports memory, CPU, and I/O stats.
- 118. **free** – Shows memory usage.
- 119. **lscpu** – Displays CPU architecture info.
- 120. **lsblk** – Lists block storage devices.
- 121. **lspci** – Lists PCI devices.
- 122. **lsusb** – Lists USB devices.
- 123. **dmesg** – Displays kernel-related messages.
- 124. **uptime** – Shows system running time.
- 125. **arch** – Displays system architecture.
- 126. **env** – Shows all environment variables.

Text Processing

- 127. **cat** – Displays the contents of a file.
- 128. **tac** – Displays contents of a file in reverse order.
- 129. **nl** – Numbers the lines of a file.
- 130. **more** – Views files one page at a time (forward only).
- 131. **less** – Advanced pager to view files forward and backward.
- 132. **head** – Displays the beginning lines of a file.
- 133. **tail** – Displays the ending lines of a file.
- 134. **cut** – Removes sections from each line of input.
- 135. **split** – Splits a file into pieces.
- 136. **paste** – Merges lines of files horizontally.
- 137. **sort** – Sorts lines in a file.
- 138. **uniq** – Removes duplicate lines from sorted data.
- 139. **wc** – Counts lines, words, characters.
- 140. **tr** – Translates or deletes characters.
- 141. **col** – Filters control characters.
- 142. **fmt** – Formats text for readability.
- 143. **fold** – Wraps text at a specified width.
- 144. **strings** – Extracts printable strings from binary files.
- 145. **grep** – Searches text using patterns.
- 146. **egrep** – Extended version of grep with more regex support.
- 147. **fgrep** – Searches fixed strings (no regex).
- 148. **awk** – Pattern scanning and text processing language.
- 149. **sed** – Stream editor for filtering and transforming text.
- 150. **xargs** – Builds and executes command lines from standard input.
- 151. **tee** – Reads from standard input and writes to file and stdout.
- 152. **rev** – Reverses lines character-wise.
- 153. **cut** – Cuts sections from each line (by delimiter or byte).
- 154. **yes** – Outputs a string repeatedly until stopped.

Shell Scripting & Variables

- 155. **bash** – GNU Bourne Again SHell, standard shell on most systems.
- 156. **sh** – Original Bourne shell.
- 157. **alias** – Creates shortcuts for commands.
- 158. **unalias** – Removes defined aliases.

- 159. **export** – Sets environment variables.
- 160. **source** – Executes a script within the current shell.
- 161. **read** – Reads input from user into a variable.
- 162. **set** – Sets shell options and positional parameters.
- 163. **unset** – Removes a variable or function definition.
- 164. **declare** – Declares variables with attributes.
- 165. **trap** – Catches signals and executes commands.
- 166. **shift** – Shifts positional parameters in scripts.
- 167. **test** – Evaluates conditional expressions.
- 168. **[]** – Alternative syntax for test.

Job Scheduling

- 169. **cron** – Time-based job scheduler.
- 170. **crontab** – Installs, lists, and removes cron jobs.
- 171. **at** – Schedules a one-time task.
- 172. **batch** – Schedules tasks to run when system load allows.
- 173. **anacron** – Runs scheduled jobs missed due to downtime.
- 174. **systemctl list-timers** – Lists all scheduled timers in systemd.

System Services (Systemd)

- 175. **systemctl** – Manages services and the systemd system.
- 176. **service** – Legacy tool to manage services.
- 177. **journalctl** – Views logs managed by systemd.
- 178. **loginctl** – Manages user logins in a systemd environment.
- 179. **hostnamectl** – Configures hostname and related settings.
- 180. **timedatectl** – Configures date and time.
- 181. **localectl** – Configures system locale settings.

Log Management

- 182. **logrotate** – Rotates and compresses log files.
- 183. **tail -f** – Follows a file (commonly used for real-time log viewing).
- 184. **less /var/log/syslog** – Views system logs page by page.
- 185. **journalctl -xe** – Views system logs with error details.

Monitoring & Performance

- 186. **iostat** – Shows CPU and I/O statistics.
- 187. **vmstat** – Reports on memory, swap, I/O, system activity.
- 188. **sar** – Collects, reports system activity.
- 189. **uptime** – Shows system load averages.
- 190. **free -h** – Displays human-readable memory usage.
- 191. **watch** – Repeats and displays output of a command periodically.
- 192. **top -n 1** – Captures system processes snapshot once.
- 193. **iotop** – Monitors I/O usage by processes.
- 194. **dstat** – Versatile resource statistics viewer.
- 195. **glances** – Cross-platform monitoring tool.
- 196. **nmon** – Performance monitoring for CPU, memory, network, and more.
- 197. **mpstat** – Shows CPU usage for each processor.

- 198. **tlload** – Shows a graph of system load average.
- 199. **uptime -p** – Shows pretty uptime format.
- 200. **hostname -I** – Shows all IP addresses assigned to the host.

Referências

DevOps Shack. **200 Linux Commands.**